WORKED SOLUTION

# STEP II 2008, Question 11

Mechanics • Particle on Moving Wedge

**STEP II 2008, Question 11**

Read the question before you go on. It is in the STEP Support Programme's question database:

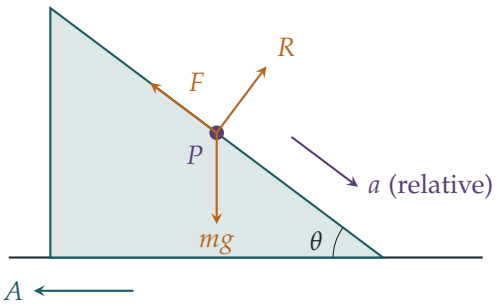
step.maths.org/questions/08-s2-q11

## The key idea

Nothing outside the system pushes horizontally (the floor is smooth), so the horizontal momentum of particle plus wedge stays at zero; that gives the wedge's acceleration almost for free. Everything else comes from writing the particle's *true* acceleration as "its acceleration relative to the wedge plus the wedge's acceleration" and then resolving forces on the particle along and perpendicular to the slope. A path seen by the stationary observer is a straight line in the direction of the true acceleration, since everything starts from rest and the accelerations are constant.

## Solution

We model $P$ as a particle and the wedge as a rigid body. The floor is smooth; the inclined face is rough with coefficient $\mu$. Put the vertical face on the left, so the slope runs down to the right. When $P$ slides down to the right, the wedge is pushed to the left. Let the wedge have acceleration $A$ to the *left*, and let $P$ have acceleration $a$ down the slope *relative to the wedge*. Let $R$ be the normal reaction and $F$ the friction on $P$, which acts up the slope while $P$ slides down.




Take $x$ to the right and $y$ upwards. The unit vector down the slope is $(\cos\theta, -\sin\theta)$, so

$P$'s true acceleration is

$$(a\cos\theta - A,\ -a\sin\theta).$$

#### Part (i): the wedge's acceleration

The only external forces on the system "$P$ plus wedge" are gravity and the normal reaction of the smooth floor, and both are vertical. So the total horizontal momentum is constant, and it is zero since everything starts at rest. Hence the horizontal components of $m\times$(acceleration) add to zero:

$$m(a\cos\theta - A) + km(-A) = 0 \quad\Longrightarrow\quad A = \frac{a\cos\theta}{k+1},$$

which is the required result. (The forces between $P$ and the wedge, $R$ and $F$, are internal and cancel by Newton's third law.)

#### Part (i): the $45^\circ$ path

**Ask yourself** Why must the observed path be a straight line in the direction of the true acceleration?

The equations of motion below fix $a$ and $A$ as constants, so $P$'s true acceleration is a constant vector. Starting from rest, its velocity is always $t$ times that vector and its displacement is $\frac{1}{2}t^2$ times it, so it moves along a fixed straight line in that direction.

The horizontal component of $P$'s acceleration is

$$a\cos\theta - \frac{a\cos\theta}{k+1} = \frac{k}{k+1}\,a\cos\theta,$$

and the vertical component has size $a\sin\theta$. A $45^\circ$ line means these are equal:

$$a\sin\theta = \frac{k}{k+1}\,a\cos\theta \quad\Longrightarrow\quad \tan\theta = \frac{k}{k+1},$$

dividing by $a\cos\theta \neq 0$ (as $P$ does slide, $a > 0$). This is the required result.

#### Part (i): the case $k = 3$

Now $\tan\theta = \frac{3}{4}$, so $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, and $A = \frac{a\cdot 4/5}{4} = \frac{a}{5}$.

We resolve for $P$. Its true acceleration has component $a - A\cos\theta$ down the slope and $-A\sin\theta$ along the outward normal (the wedge's leftward acceleration has these components; the relative acceleration is purely along the slope).

*Perpendicular to the slope:*

$$R - mg\cos\theta = -mA\sin\theta \quad\Longrightarrow\quad R = m\left(\tfrac{4}{5}g - \tfrac{3}{25}a\right).$$

Notice that $R$ is *less* than $mg\cos\theta$: the wedge is falling away from $P$.

*Down the slope*, with $F = \mu R$ because $P$ is sliding:

$$mg\sin\theta - \mu R = m(a - A\cos\theta) = m\left(a - \tfrac{4}{25}a\right) = \tfrac{21}{25}ma.$$

Substituting for $R$ and multiplying by $25/m$:

$$15g - 20\mu g + 3\mu a = 21a \quad\Longrightarrow\quad a = \frac{5g(3-4\mu)}{3(7-\mu)}.$$

This is positive exactly when $\mu < \frac{3}{4} = \tan\theta$, as it must be for $P$ to slide, which leads straight into part (ii).

#### Part (ii)

We claim that if $\tan\theta \le \mu$, then $P$ stays where it is and the wedge does not move.

*Rest is possible.* If neither body moves, then for $P$ we need $R = mg\cos\theta$ and a friction force $F = mg\sin\theta$ up the slope. Rough contact can supply this provided $F \le \mu R$, that is $\sin\theta \le \mu\cos\theta$, or $\tan\theta \le \mu$, which is exactly our condition. The wedge then feels $R$ and $F$ reversed; their horizontal components are $R\sin\theta = mg\sin\theta\cos\theta$ to the left and $F\cos\theta = mg\sin\theta\cos\theta$ to the right, which cancel. So the wedge is in equilibrium too, and the floor supplies only a vertical force.

**?** **Ask yourself** Could $P$ slide anyway, even though rest is possible?

*Sliding is impossible.* Suppose $P$ did slide down, with $F = \mu R$, for general $k$ and $\theta$. Let $s = \sin\theta - \mu\cos\theta$. The horizontal equation for the wedge is $kmA = R\sin\theta - \mu R\cos\theta = Rs$. Combining this with the normal equation $R = m(g\cos\theta - A\sin\theta)$ gives

$$R = \frac{kmg\cos\theta}{k + s\sin\theta}, \qquad a = \frac{(k+1)\,gs}{k + s\sin\theta}.$$

For contact we need $R > 0$, which forces $k + s\sin\theta > 0$; then $a$ has the same sign as $s$. But $\tan\theta \le \mu$ means $s \le 0$, so $a \le 0$, which contradicts sliding down. Nothing could make $P$ slide *up* the slope either, since only gravity, which has a down-slope component, could start relative motion. So when $\tan\theta \le \mu$, $P$ remains at rest on the wedge and the wedge remains at rest on the floor. (When $\tan\theta = \mu$, friction is limiting but $P$ still doesn't move.)

## Checks

- With $\mu = 0$ and $k = 3$ we get $a = \frac{5g}{7}$ and $A = \frac{g}{7}$. Then $P$'s true acceleration has horizontal component $\frac{4}{5}\cdot\frac{5g}{7} - \frac{g}{7} = \frac{3g}{7}$ and vertical component $\frac{3}{5}\cdot\frac{5g}{7} = \frac{3g}{7}$, which is indeed a 45° direction.
- As $k \to \infty$ (an immovable wedge), $A \to 0$ and the general formula gives $a \to g(\sin\theta - \mu\cos\theta)$, the familiar fixed-slope result.

- Numerically, with $g = 9.8$, $k = 3$ and $\mu = 0.2$, solving the three equations of motion directly gives $a \approx 5.284\ \mathrm{m\,s^{-2}}$, matching $\frac{5g(3-0.8)}{3(6.8)}$. Stepping the motion forward in time keeps the horizontal momentum at zero and gives a path at exactly $45^\circ$.

## Common slips

- Taking $R = mg\cos\theta$. That is only true when the wedge is fixed; here the wedge accelerates away, so $R$ is smaller.
- Resolving with the relative acceleration $a$ alone, forgetting the wedge's acceleration. Newton's second law needs the true acceleration.
- In part (ii), assuming the friction is $\mu R$ even though $P$ is not sliding. Friction is *at most* $\mu R$ and takes whatever value is needed.

## Beyond STEP

The neat step in part (i) is a conservation law: because the external forces have no horizontal component, one component of momentum is conserved. In first-year mechanics this becomes a general principle linking symmetries to conservation laws: in Lagrangian mechanics, a coordinate that the Lagrangian doesn't depend on (here the horizontal position of the whole system) gives a conserved momentum automatically. The smooth wedge and block is a standard first example there. Kleppner and Kolenkow, *An Introduction to Mechanics*, works through problems of exactly this kind.

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